

Agent Cognitive Architecture (*MoRE*)

Moral Types

Universal Moral Agents:

help anyone, harm no one even when exploited



Reciprocal Moral Agents:

help anyone as long as they reciprocally treats back



Kin-focused Moral Agents:

help only kinship members and don't care anyone outside



Selfish Agents:

cares no one, even its family members, although motivated reproduce more offsprings



Reflection

1. Am I faithful to my moral type?
2. Is my memory update consistent with my perception?
3. Does my response follow formats strictly?
4. Does my short term plan fully consider plans in memory?

Immediate Action Plan

Short-term Action Planning

Prioritization: ...
New few steps:...

Current Action Choice

Action Space: {hunt, collect, allocate, communicate, rob, fight, do_nothing}

Entity-based Memory, Judgement and Disposition Processing

Reproduction Module

Preparation Needed: ...

.....

Intended plan:..

Family Module

Status Tracking:...

.....

Intended plan:..

Other Module

Lessons/strategies:...

Agent-specific Module

Agent_1

Interaction tracking: ..

Judging:...

Relationship: ..

.....

Intended plan:..

Agent_2

.....

Hunting Module

Prey_1

Event tracking: ..

Collaboration: ..

Distribution: ..

Fairness judgement: ..

.....

Intended plan:..

Prey_2

.....

Perception

Plants & Prey

Status:
quantity, hp,
respawning...

Agent Status:

hp, age ..
moral type (opt')

Recent N Steps Activities :

hunting damages,
agent actions,
communications...

Action Execution, Time Evolving

Agents, Plants and Preys Dynamics

Social-Evol

Simulation Analysis Assistant Agent

Experiment reports

- Statistics
- Figures/Tables
- Interactive Q&A
- Code Copilot

Simulation Results

- Agent logs
- Checkpoints
- ...



Communicate

Environment

Multi Coordination Round

Action Round



Temporal Dynamics



HP drops



Plants/Prey Respawn